

1. Mean, Median, Mode – Finding the Center

We started with **measures of central tendency** — the *mean*, *median*, and *mode*.

- **Mean** is the average — great when the data is clean and balanced.
- **Median** is the middle value — perfect when your data has outliers.
- **Mode** is the most frequent value — often useful for categorical features.

💡 *In ML, using the median instead of mean can make your model more robust to extreme values.*

2. Variance and Standard Deviation – Measuring the Spread

Then we explored **how spread out the data is**.

- **Variance** tells us how far data points are from the mean, on average.
- **Standard deviation (σ)** is just the square root of variance, making it easier to interpret in the same unit as the data.

🔗 *In ML, features with large variance can dominate the model, which is why we normalize or standardize data before training.*

3. Percentiles, Quartiles, IQR, and Z-score

Next, we moved into **relative positioning and outlier detection**.

- **Percentiles** and **quartiles** divide data into ranks or quarters.
- **Interquartile Range (IQR)** = $Q3 - Q1$ — helps identify the “middle 50%” of data.
- **Z-score** measures how far a point is from the mean in terms of standard deviations.

💧 *ML use case: Z-scores are often used for anomaly detection or data cleaning steps.*

4. Distribution Shapes and Outliers

We learned how the **shape of the distribution** tells a story:

- **Symmetric** distributions like the normal curve are balanced.
- **Right-skewed** distributions (like income) have long tails to the right.
- **Left-skewed** distributions tail off to the left.

Outliers stretch the tails — and using the **IQR method**, we can set boundaries to detect them.

💬 *In ML, outliers can drastically affect models like Linear Regression, so detecting and handling them is critical.*

5. When to Use Median and IQR

Finally, we saw **when median and IQR beat mean and standard deviation** — that’s when data is **not symmetric**, or has **outliers**.

For example, in predicting **income**, a few millionaires can ruin the mean, but the median stays honest.

✅ *So in ML preprocessing, median and IQR are often used in robust scaling.*

Conclusion

So that wraps up Module 1!

You now understand how to describe your data — its **center, spread, and shape** — and how to detect unusual patterns before any modeling begins.

In the next module, we'll move from describing data to **understanding probability and Bayes' theorem**, which will build the foundation for how ML actually "learns."